The issue becomes harder when we get to 3 parameters: one workaround is to introduce “time” as the fourth dimension, and as such the plot can be shown as a 3D wave that changes depending on the third parameter.

Besides being hard to make sense of, this solution doesn’t solve the problem when we have 4 or more parameters. What we can usually do in these cases is locking k-2 parameters, out of k total variables, and see how the function behaves as the other 2 variables change.

We’ll do something similar with our graph embedding problem, to show you how this works, and understand better our problem.

But first of all, what function are we talking about in this case?

16.2.1 Cost Functions

We call the target of our optimization a **cost function**: it expresses well the idea that our solution (each solution we try) has a cost, and we are trying to minimize it.

There is a vast category of problem called **optimization problems** for which finding a solution is equivalent to exploring the problem space and eventually picking the solution with the lowest cost; for these problems usually there are many possible definitions of cost functions that can be used: some are better than others (we’ll see why), and so – when we do have a choice – it’s important to spend time and choose wisely, because it can have a great impact on how fast we can find an optimal solution, or sometimes even on whether we can find one. In general, however, most optimization problems are proven to be NP-hard, regardless of the specific choice of cost function.